Getnera: Safety, Billing and Backup Foundation (Developer Brief)

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Purpose and decisions

Build Getnera so that WhatsApp cannot restrict it, customers pay in a controlled way, and one server failure cannot lose data. Do the items in section 6 in order, one small pull request each, without stopping current work.

Decisions already made:

- **Meta billing:** each customer pays Meta directly for WhatsApp messages, using their own WhatsApp Business Account (WABA). Do not build re-billing of Meta fees yet.
- **Wallet:** covers AI replies and extras only (extra seats, extra numbers).
- **One WABA per customer.** Never send external customers' traffic from Afra's own number.
- **Plan prices and wallet rates below are drafts.** Kamar confirms them after measuring the real AI cost per reply.
- **Reseller mode** (re-billing Meta fees with a margin) waits until Getnera is a Meta Tech Provider and has 20 or more paying customers.
- **Source of the findings:** the LeadOS and Nera WhatsApp reuse map (PR #6 in nera-sass).

Services and features

Getnera sells one outcome to small businesses in Tamil Nadu and Puducherry: every customer message answered, nothing missed, bookings captured. The software has five modules, and a done-for-you service sits on top.

Module	What it does	Plans
Shared inbox	One WhatsApp number answered by a team: assign chats, internal notes, quick replies, labels, working hours, away messages	All

Module	What it does	Plans
CRM	Contacts with tags, notes, source and history; simple pipeline (new, interested, booked, paid, lost); CSV import and export	All
AI replies	Answers from the business's knowledge base in English, Tamil or mixed; hands over to a person when unsure or asked	WhatsApp, Complete
Booking	Captures name, service, date, time and contact; confirms; writes to Google Calendar and Sheets; sends reminders	WhatsApp, Complete
Follow-ups and reminders	Automatic follow-up after no reply, appointment and payment reminders on a schedule	WhatsApp, Complete
Campaigns	Template broadcasts to opted-in segments, with cost preview and results (next section)	Complete, limited on WhatsApp
Reports	Chats, response time, bookings, leads by source; weekly summary to the owner by WhatsApp or email	Basic, Standard, Full by plan
Team and access	Roles (owner, manager, agent), per-user access, activity log	All

Must work before the first paying customer. These are known open problems from live tests and are release blockers, not polish:

- The AI must not re-greet with a full recital on every message. Open with a light hello that mirrors the customer.
- Once a brand or topic is detected it stays sticky and switches only when the customer clearly changes topic.
- The booking flow must actually capture date, time and contact, confirm, and trigger the action. Booking fields currently do not persist.
- The brand must not flip between messages on a multi-brand number.

Brand voice per customer. Each tenant has its own tone, offers, FAQs, banned words and seasonal focus, stored as a knowledge base that every AI reply reads first. Thedal's "Don's Brain" is the same idea; build one shared component.

Human control. The owner can pause the AI per chat or for everyone, review the AI's replies in a log, and set topics it must always pass to a person (complaints, refunds, medical advice).

Services, not software (the paid extras):

- **Done-for-you setup** (one-time Rs 2,999 to Rs 4,999): connect WhatsApp, add the Meta payment method, build the knowledge base in Tamil and English, submit the first templates.
- **Template writing** so that Meta approves them the first time.
- **Staff training** (one hour, on video or in person in Puducherry).
- **Managed monthly service** for customers who want Getnera's team to run their campaigns and reports. Price it separately from the software.

WhatsApp safety: prevent restrictions

Meta restricts numbers mainly when people report or block messages, or when businesses send without permission. Every item below is a requirement.

1. **Opt-in for every contact.** Store when, how and for what purpose the contact agreed (form, QR code, chat). Refuse marketing sends to anyone without a record. Never message bought or scraped lists.
2. **Opt-out.** A STOP or unsubscribe reply blocks marketing to that contact immediately, in every tenant campaign.
3. **Approved templates only** outside the 24-hour customer service window. Free text only inside it.
4. **Frequency caps.** A per-contact weekly limit on marketing messages, set per tenant. No sending one message to a very large list at once.
5. **Warm-up.** A new number starts with small volumes to engaged contacts, and the daily limit rises in steps.
6. **Per-tenant send queue with rate limiting,** so one customer's burst cannot trigger Meta limits for others.
7. **Quality watch.** Read each number's quality rating and messaging limit through the API on a schedule or webhook. If the rating drops, **pause that tenant's campaigns automatically** and alert the admin.
8. **Official WhatsApp Business API only.** No unofficial libraries and no personal-number automation.
9. **Signed webhooks.** Verify `X-Hub-Signature-256` on every Meta webhook and fail closed if the app secret is unset (the LeadOS fix is PR #244).

10. **No platform fallback.** Remove every fallback to the global token or phone number (`followups.js:189`, `templates.js:195`). A tenant without credentials gets a clear error, never a send from the platform number.
11. **Encrypt each tenant's WhatsApp credentials at rest,** and never write raw webhook payloads to disk or to a publicly served folder.
12. **Business verification.** Complete Meta Business Verification for Afra Ventures before the first paying customer, since verified businesses get higher limits.
13. **Embedded Signup and Tech Provider.** The current Embedded Signup is a placeholder that trusts browser-supplied IDs. Build it properly and start the Tech Provider and App Review application now, because approval takes weeks.
14. **Content rules.** No misleading text, shouting or off-topic content, and each brand's messages stay on its own topic.
15. **Review requests.** Ask every customer for a review, not only the happy ones. Selective asking counts as review gating on Google.

If WhatsApp restricts or blocks a number

The system must keep serving customers while the appeal runs, and must never use a second number to dodge a ban.

- **Appeal evidence ready.** For every tenant, store opt-in records, approved templates and message logs in a form that can be exported as one file within minutes.
- **Backup number.** Support registering a second number under the same verified business, with templates approved and tested. It is a continuity measure for the same legitimate business, not a way to re-register a banned number.
- **Fallback channels.** If WhatsApp sending fails or is paused, the system sends the same message by email, and by SMS or voice for urgent ones (through a DLT-registered provider). Thedal's alerts already have the email fallback in PR #235; reuse that pattern.
- **Own the data.** Contacts, consent records and chat history live in the database and export to CSV at any time, so no platform can lock them away.
- **Customer notice.** A ready status message and a one-page notice for customers, so they hear about an outage from Getnera first.
- **Clear the live warning.** The LeadOS logs show `Business Account locked (#131031)` for two phone numbers. Kamar checks WhatsApp Manager for the cause; the developer does not touch these numbers.

Start here: plans and wallet in plain words

Build this in two steps. Step 1 is enough for the first pilot customers.

Step 1: plans with limits (no wallet yet).

- A **plan** is a monthly subscription with limits: seats, WhatsApp numbers, contacts, AI replies included, campaigns on or off.
- Store plans in a `plans` table. Each tenant has one row in `subscriptions` that points to a plan and shows its status and renewal date.
- Before any gated action, check the limit. If it is exceeded, show a clear message and an upgrade button. Do not charge anything extra and do not block existing contacts or chats.
- Pilot customers pay nothing or pay by a Razorpay payment link sent by Kamar. Subscription automation comes after.

Step 2: wallet (after the pilots).

- The **wallet** is a prepaid balance. It is used only for things beyond the plan: extra AI replies, extra seats and extra numbers.
- A customer adds money by Razorpay. Each top-up is one row in `wallet_ledger`. Each charge is one row with a negative amount. The balance is the sum.
- If the balance is too low, block only the paid extra and show a top-up button. Replies to customers inside 24 hours keep working.

What you do not build: Meta's message fees. The customer pays Meta directly on their own WhatsApp account, so Getnera never bills or refunds them.

Worked example. Nera WhatsApp includes 500 AI replies a month. The customer uses 620. The first 500 are covered by the plan. The extra 120 are charged from the wallet at the rate in the pricing table. If the wallet is empty, replies beyond 500 stop being AI-generated and go to a human, with a top-up prompt to the owner.

Questions answered: Who pays Meta? The customer. What are the prices? The draft table below, kept in data so they can change without code. Do we need refunds? Only a manual admin adjustment row. Do we need invoices? Yes, a GST invoice per payment, which Razorpay and a small invoice template can produce.

Plans and wallet

Subscriptions pay for the software. The wallet pays for usage beyond the plan. Limits and rates live in tables, never in code.

Draft plans (Kamar to confirm; the prices sit near Indian competitors such as Chakra Chat from Rs 999 and WATI Growth from Rs 2,499, and need checking against the real AI cost per reply):

Plan	Price per month (Rs)	Seats	WhatsApp numbers	Contacts	AI replies included	Campaigns	Reports
Nera Desk	799	3	1	1,000	0	No	Basic
Nera WhatsApp	1,999	5	1	5,000	500	Limited	Standard
Nera Complete	3,999	10	2	25,000	2,000	Yes	Full

A contact is one stored phone number with its chat history. At the limit, stop adding new contacts after a short grace period with a warning, and never delete or block existing ones.

Wallet rules

- **Append-only ledger.** `wallet_ledger`: tenant_id, type (topup, ai_reply, extra_seat, extra_number, refund, adjustment), amount_paise, ref_id UNIQUE, meta, actor, created_at. No updates or deletes. Balance is the sum of the rows.
- **Top-up through Razorpay** (UPI, cards, netbanking). Credit only after verifying the webhook signature, and make it idempotent on payment_id. Minimum Rs 500, maximum Rs 50,000.
- **Debits** for AI replies beyond the plan and for extras, at rates from a pricing table. Use a transaction with row locking so concurrent debits can never take the balance below zero.
- **Reserve and settle** for any bulk action: reserve the estimate, settle the actual cost, release the difference.
- **Insufficient balance** blocks the paid action with a clear message. Replies to customers who messaged within the last 24 hours are never blocked.
- **Alerts** at 20% of the last top-up and at a fixed floor, by email and in-app. Optional auto top-up with a saved mandate.
- **Subscriptions:** Razorpay with UPI AutoPay or e-mandate for recurring payment, and a GST invoice (18%) for every payment and top-up.
- **Tenant isolation.** Every wallet and ledger query is scoped by tenant_id. Admin adjustments need a reason and are logged with the actor.
- **Terms.** The wallet is non-withdrawable, non-transferable and usable only for Getnera services. Kamar confirms refund, expiry and GST wording with the CA before launch.

- **Campaign cost preview.** Before a campaign sends, show the expected Meta cost (paid by the customer to Meta), since indicative India rates are about Rs 0.86 for a marketing message and Rs 0.115 for utility, and customers may blame Getnera for a surprise bill.

Campaigns and templates

Campaigns are where a customer's number is most at risk, so every feature below includes its safety rule. WhatsApp only lets a business start a conversation with a pre-approved template. Replies inside 24 hours of the customer's last message need none.

Campaigns, version 1

- Audience from tags or segments. Contacts without a recorded opt-in are excluded automatically.
- Pick an approved template and fill its variables per contact.
- **Cost preview** before sending, since the customer pays Meta directly. Indicative India rates are about Rs 0.86 for marketing and Rs 0.115 for utility, and must be checked against the current rate card.
- Schedule a time. Sending is limited to daytime hours.
- Send in batches through the per-tenant queue, with a lower daily cap for new numbers.
- Results: sent, delivered, read, replied, failed, opted out.
- Safety: frequency cap per contact, automatic STOP handling, and an automatic pause if the number's quality rating drops.

Later: drip sequences, A/B testing of two templates, click-to-WhatsApp ad tracking, birthday and renewal reminders, and revenue attribution (campaign to booking).

Templates

- **Categories:** marketing, utility and authentication. Meta decides the category. A promotion submitted as utility may be reclassified, which changes the price.
- **Variables** such as `{{1}}` must match the count the code sends, or Meta rejects the send.
- **Languages:** approve English and Tamil versions of each template and send the one that matches the contact.
- **Create and submit through the API,** and show each template's status (pending, approved, rejected) with Meta's reason inside Getnera. Approval takes minutes to a day.
- **Quality:** show each template's quality and disable one automatically if Meta pauses it.

- **Default flow:** Getnera submits templates for the customer from the starter library, and the customer edits the text. Small businesses often write templates that Meta rejects.
- Every marketing template carries an opt-out line.

Starter library (to confirm which industries first: clinics, schools and coaching, shops and services)

Use	Category	Example, shortened
Appointment confirmation	Utility	Hi {{1}}, your appointment at {{2}} is confirmed for {{3}}.
Appointment reminder	Utility	Reminder: your visit is tomorrow at {{1}}. Reply 1 to confirm.
Payment or fee reminder	Utility	Hi {{1}}, Rs {{2}} is due on {{3}}. Pay here: {{4}}
Order or booking update	Utility	Your order {{1}} is {{2}}.
Offer or festival greeting	Marketing	Happy {{1}}! This week: {{2}}. Reply STOP to unsubscribe.
Review request	Utility or marketing	Thanks for visiting {{1}}. How was it? {{2}}

Backups and resilience

The Hostinger server runs about 20 apps behind one Postgres container, and port 5432 is open to the internet. The current backup state has not been checked yet, so confirm it first (see section 6).

1. **Automatic daily database backups** with `pg_dump` from the Postgres container, kept 14 days, plus weekly and monthly copies. Store them off the server, in another provider's storage, encrypted.
2. **Restore test every month** on a spare database, with the result written down. A backup that has never been restored is not a backup.
3. **VPS snapshots** weekly and before every deployment.
4. **Secrets.** Keep `.env` files and keys in a password manager or encrypted vault, never in the repository, a chat or a public folder. Rotate any value that has ever appeared in a repository.

5. **Close port 5432.** First switch the apps to connect on `127.0.0.1`, then block the port with a firewall that allows only SSH and web traffic.
6. **Isolation.** A separate database user per app, with no sharing. Kanakku and LeadOS shared one user until the 4 November rotation. Later, consider a separate server for any product sold to customers.
7. **Monitoring.** An uptime check on each API, and alerts on pm2 restarts. `getnera-api` and `npp-plot-staging-api` each showed 140 or more restarts in 9 hours, so find the cause.
8. **Recovery runbook.** A one-page document covering who does what, where the backups are, and how to restore, with a target of back online within 2 to 4 hours.
9. **Public folders.** Check that no served folder (for example `/uploads`) contains logs, exports or tokens.

Build order and checklist

Safety and backups come before billing, because a blocked number or a lost database costs more than a late wallet. One small pull request per item.

1. Webhook signature check, with no platform fallback and no raw payload logs (section 2, items 9 to 11).
2. Per-tenant encrypted WhatsApp credentials and tenant scoping on every query.
3. Opt-in records, opt-out handling and the 24-hour window rule.
4. Per-tenant send queue, rate limits and frequency caps.
5. Quality-rating watch with automatic campaign pause and an admin alert.
6. Off-server database backups and a tested restore. Then close port 5432.
7. Plans table, plan limits and Razorpay subscriptions with GST invoices.
8. Wallet ledger, top-up and debits, with reserve and settle.
9. Email, SMS or voice fallback when WhatsApp is paused.
10. Embedded Signup built properly, then the Tech Provider and App Review application.

Tests for each item: bad or missing signature, duplicate payment webhook, concurrent debits, insufficient balance, tenant isolation, opt-out respected, and a paused campaign after a quality drop.

Tasks for Kamar (not the developer)

- [ ] Open WhatsApp Manager and find the cause of the `#131031` Business Account lock.
- [ ] Complete Meta Business Verification for Afra Ventures.
- [ ] Confirm the three plan prices after the developer measures the real AI cost per reply over 100 real conversations.

- [ ] Check the wallet, refund, expiry and GST wording with the CA.
- [ ] Check the server's current backup state in the Hostinger terminal and send the result: `crontab -l; ls -la /root/backups /var/backups 2>/dev/null | head -20; docker ps --format '{{.Names}} {{.Image}}'`
- [ ] Fix the Razorpay key on Kanakku, which logs `Authentication failed`, before Getnera uses the same account.

Open questions

- Does Meta accept UPI or netbanking for WhatsApp billing for Indian businesses, or only cards? Verify before promising self-service setup.
- Should the wallet price include GST or add it at checkout?